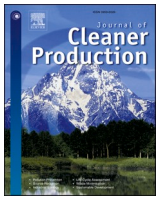
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ESGReveal: An LLM-based approach for extracting structured data from ESG reports

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ABSTRACT

As an important source for disclosure a company's environmental, social, and governance (ESG) performance, stock exchanges gradually strengthen their requirements for listed companies to periodically submit their ESG exports. However, these documents are often unstructured, making it difficult to directly evaluate a company's disclosure level as well as the performance quantitatively. In this study, we develop a quantitative framework, ESGReveal, for assessing corporate ESG performance based on large language model (LLM) techniques. Specifically, by integrating retrieval-augmented generation (RAG) technology with LLMs, we extract relevant performance data from complex corporate ESG reports. The ESGReveal framework consists of three primary modules: an ESG Metadata module for standardized queries, a Report Preprocessing module for database construction, and an LLM Agent module for data extraction. We evaluated the performance of various LLMs, including GPT-3.5, GPT-4, ChatGLM, and QWEN, and found that GPT-4 achieved 76.9% accuracy in data extraction and 83.7% accuracy in disclosure analysis, showing the best improvement over baseline models. We applied this ESGReveal model to 2249 ESG reports published by 166 companies across 12 industries listed on the Hong Kong Stock Exchange (HKEx), analyzing the disclosure and performance of key ESG indicators. Results show that for mandatory environmental and social indicators required by HKEx, the sample companies achieved disclosure rates of 69.5% and 57.2%, respectively. Different industries exhibited varying performance in key ESG indicators, such as the proportion of direct and indirect greenhouse gas emissions, highlighting key areas for future emission reduction efforts. These findings underscore the need to strengthen ESG practices across sectors and emphasize both general and sector-specific ESG initiatives. In summary, by leveraging the capabilities of LLM and RAG technologies, ESGReveal offers a practical and efficient solution to the pressing need for consistent and accurate ESG information retrieval.

1. Introduction

The concept of Environmental, Social, and Governance (ESG) has gained significant traction in modern corporate practices, emphasizing the importance of non-financial factors in assessing business performance. This trend is supported by a growing body of literature that highlights the materiality of ESG issues to long-term business success (Tsang et al., 2023). There is a growing demand from stock exchanges, such as the Hong Kong Stock Exchange (HKEx), for listed companies to disclose their ESG performance, reflecting a global shift towards sustainability. This demand is driven by the recognition that ESG disclosures are critical for investors to make informed decisions and for companies to demonstrate their commitment to sustainable practices (Bui et al., 2020). ESG disclosures are also crucial for corporate sustainable development and provide valuable references for investors, regulatory bodies, and other stakeholders to make informed decisions

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(Doe, 2021).

A critical issue in climate action transparency is ensuring that companies' reported efforts to reduce carbon emissions are genuine and effective. This challenge is addressed in recent studies that focus on the verification and validation of corporate ESG claims (Abhayawansa and Tyagi, 2021; Liu et al., 2024). It is essential to compare the sustainability performance across different industry sectors to identify those with higher environmental and social risks that may require more attention or regulation. Such comparisons are necessary to understand the variability in ESG performance and to inform targeted interventions (Clarkson et al., 2019).

The primary challenge in ESG quantitative research is the reliance on self-reported, unstructured ESG reports by companies, which complicates the process of quantifying disclosure levels and ESG performance. This issue is further complicated by the lack of standardized reporting frameworks and the diversity of reporting practices across companies (Fischbach et al., 2022). Existing research has addressed some aspects of ESG analysis, but there are still limitations in data extraction and standardization methods that hinder regulatory and analytical requirements. These limitations include the inability to process large volumes of unstructured data and the lack of tools to extract and analyze ESG information accurately (Pasch and Ehnes, 2022). There is still a gap particularly evident in the lack of detailed analyses of ESG disclosures across different industries and the absence of benchmarks for ESG performance (Schiemann and Tietmeyer, 2022).

This study introduces ESGReveal, a novel solution designed to address the existing problems and technological shortcomings. By employing advanced technologies and extensive data, we have automated the analysis of ESG disclosure levels and core data extraction from a significant number of companies, thereby optimizing the ESG data quantification process. This approach aims to fill the gap identified in the literature by providing a systematic and automated method for ESG data analysis. The integration of Large Language Models (LLM) and Retrieval Augmented Generation (RAG) techniques in ESGReveal enhances the efficiency and accuracy of extracting key information from unstructured ESG reports. This innovative approach leverages the latest advancements in natural language processing to improve the precision of ESG data extraction and analysis (Lewis et al., 2020). This research contributes to enhancing the transparency and standardization of corporate ESG data, providing a more reliable analytical tool for regulators, investors, and researchers. The findings reveal differences in ESG disclosures and performances across industries, offering insights for future emission reduction efforts and guiding the sustainable development of various sectors.

The paper is structured as follows: Section 1 introduces the concept of ESG and the challenges associated with ESG data extraction; Section 2 reviews related work in the field; Section 3 describes the methodology and the components of ESGReveal and the datasets used and the experimental setup; Section 4 details the results of the accuracy assessment, LLM benchmarking, and ablation study; Section 5 discusses the implications of the findings; and Section 6 concludes the paper, summarizing the key contributions and outlining potential directions for future research.

2. Literature review

Amid the increasing complexity of ESG disclosure and the rapid expansion in information volume, researchers have started exploring the potential of natural language processing (NLP) techniques to enhance the precision and efficiency of ESG data extraction and analysis. Relevant research over the past five years is also listed in Supplementary Table 1, detailing author, published year, application areas, methodologies, and primary findings. Li et al., (2024) employed a combination of data mining and the Analytic Hierarchy Process (AHP) to enhance the efficiency and accuracy of ESG management practice evaluation in manufacturing enterprises. Ruberg et al. (2023) utilized the BERT architecture to automatically classify corporate ESG reports content relevant to the GRI standards, increasing the analytical efficiency for ESG assessments at the Brazilian Development Bank. Perazzoli et al. (2022) conducted an extensive analysis of 55,000 publications on ESG-related topics using NLP techniques. Luccioni et al. (2020) developed the ClimateQA model based on NLP technology to analyze information related to climate change in financial reports. Raman et al. (2020) detected ESG trends by analyzing corporate earnings call transcripts and parsing the linguistic structure within ESG texts. Fischbach et al. (2022) created a tool named ESG-Miner, which automatically extracts ESG-related information from media headlines and calculates ESG scores for companies. In these studies, advanced NLP models like BERT have demonstrated substantial progress in understanding ESG contexts and semantics (Bronzini, 2023; Lee et al., 2022; Liu et al., 2023). However, yet there remain notable limitations in areas such as adaptability to the ESG sector, the creation of custom datasets, deep information mining, and multilingual and cross-cultural adaptability (Pasch and Ehnes, 2022; Fischbach et al., 2022).

Recent advancements in natural language processing, particularly LLMs such as GPT-3.5 (OpenAI, 2023b) and GPT-4 (OpenAI, 2023a), have further enhanced this field. To address challenges related to lengthy contexts and data currency in LLM, researchers have integrated RAG paradigm (Lewisetal., 2020) with LLM, improving response accuracy with dynamic information retrieval. Current research reveals the substantial potential of LLMs in identifying sustainability goals and performance indicators within ESG reports (Zhang and Zhang, 2023; Burnaev et al., 2023). For instance, Kim et al. (2023) used ChatGPT to summarize the economic utility of corporate disclosures and revealed the link between information ("bloat") and adverse capital market consequences. Ni et al. (2023) developed the CHATREPORT system based on LLM, which analyzed 1015 corporate sustainability reports using the Task Force on Climate Related Financial Disclosures (TCFD) framework (https://www.fsbtcfd.org/) and assessed their compliance. Bronziniet al. (2023) extracted semantically structured ESG-related information from sustainability reports using LLM, unveiling intercorporate ESG action correlations. Moodley and Telukdarie (2023) enhanced the ability to identify green claims and detect "green washing" by training LLM with sustainability related textual corpora. Yang et al. (2023) demonstrated the potential of their FinGPT in recognizing ESG related information. The evolution of LLM has greatly optimized text processing methods, making the transformation of unstructured documents in ESG reports into structured data a feasible reality (Visalli et al., 2023).

Building on the foundation of existing research, this paper presents ESGReveal, a novel methodology that focuses on the extraction of data from ESG reports with a particular emphasis on the structured retrieval of numerical information. Our approach harnesses the sophisticated text processing capabilities of LLMs within RAG paradigm to systematically decompose both numerical and textual information contained in ESG reports from various standard frameworks. This systematic analysis facilitates deeper insights into ESG disclosure practices. Furthermore, in recognition of the pivotal role played by the Hong Kong Stock Exchange (HKEx) in advancing ESG initiatives in China, this study employs ESGReveal to evaluate the ESG reports of companies listed on the HKEx. Through this application, we aim to analyze the prevailing state of industry transparency and accountability in ESG, contributing towards a clearer understanding of the ESG landscape within the region.

3. Materials and methods

3.1. Data preparation: ESG reports from HKEx

We collected 2249 ESG reports in year 2022 from listed companies on HKEx (https://www.hkex.com.hk/), which are categorized into 12 sectors according to the Hang Seng Industry Classification System (Hong Kong Exchanges and Clearing Limited, 2023b). As processing all ESG

reports entailed a significant computational workload, we endeavored to select a representative sample of 166 companies, balancing industry diversity and market capitalization, for evaluating ESGReveal and performing disclosure analysis (Table 1).

3.2. Method: the ESGReveal framework

Fig. 1 illustrates a three-module design of our ESGReveal framework: 1) the ESG Metadata module, establishes a query framework that leverages ESG criteria and expertise to analyze reports; 2) the Report Preprocessing module imports ESG reports and processes them to build a database for information retrieval; 3) the LLM Agent module accesses the database built from the Report Preprocessing module to retrieve information, and engages the LLM for data extraction based on the criteria built by the ESG Metadata module. Generally, in the ESGReveal workflow, an ESG report is firstly processed by the Report Preprocessing module and then analyzed by the LLM Agent module, resulting in data that adheres to the ESG specifications defined by the ESG Metadata module.

3.2.1. ESG metadata module

The ESG Metadata module is structured to comply with international ESG standards, such as GRI and SASB, and is organized into three key components: Entities, Extensions, and Expressions, as depicted in Fig. 2. Entities outline the attributes of key performance indicators (KPIs) required to be disclosed in the ESG reports, including category, description, subcategories, and units of measurement. Extensions incorporate <Knowledge> and <SearchTerm> to adapt to various ESG compilation standards (for details see Table 2). Expressions concentrate on crafting Prompt templates and formatting the output, which allow for the automatic generation of prompts to trigger the LLM. In this study, based on the KPIs defined by the Supplementary 27 Environmental, Social and Governance Reporting Guide (Hong Kong Exchanges and Clearing Limited, 2023b) and the Supplementary 14 CORPORATE GOVERNANCE CODE (Hong Kong Exchanges and Clearing Limited, 2023a) from HKEx, we construct the Entities and Extensions in the ESG Metadata module. Adhering to the HKEx guidelines, the framework comprises 70 KPIs, divided as 12/18/4 numerical and 14/15/7 textual indicators across the E/S/G categories, totaling 34 numerical and 36 textual indicators (for details see Supplementary Table 2). Examples of Entities and Extensions for sample KPIs can be found in Supplementary Table 3.

(1) Entities

The Entities aim to simplify complex ESG issues into the format: <Aspect, KPI, Topic, Quantity>, with "KPI" denoting key performance indicators. Here, two examples of Entities for querying numerical and textual indicators on the "A1. Emissions" issue are as follows respectively.

Example 1. On the "A1. Emissions" issue, for querying total and intensity of waste emissions, <Aspect, KPI, Topic, Quantity > are "A1. Emissions", "Total waste produced (in tonnes) and, where appropriate, intensity", "Non-hazardous Waste, Hazardous Waste." and "Absolute Values".

Example 2. On "A1. Emissions" issue, regarding key actions on emissions, <Aspect, KPI, Topic, Quantity > are "A1. Emissions", "Emissions target(s) and steps taken to achieve them", "Waste, Exhaust/ Greenhouse Gases", and "Key Actions".

(2) Extensions

To enhance adaptability across diverse ESG reporting standards, we extend two components integrated into the ESG Metadata module, i.e., <Knowledge> and <SearchTerm>. The <Knowledge> component, containing domain knowledge curated by ESG experts and user friendly explanations of indicators produced by sophisticated LLM, is embedded into LLM prompts via In Context Learning to accurately carry out indicator extraction tasks. The <SearchTerm> component includes customized retrieval keywords gathered, for instance, from an extensive array of ESG reports based on the HKEx guidelines to improve the performance of the RAG retrieval.

(3) Expressions

Expressions are designed to streamline the output formatting and automate the generation of prompts that trigger the LLM. This component focuses on creating structured templates for data presentation and query formulation. The Expressions are crucial for translating the complex ESG indicators into a format that is both machine-readable and user-friendly. They are meticulously crafted to ensure that the prompts generated are clear, concise, and aligned with the ESG criteria outlined by standards such as GRI and SASB.

3.2.2. Report Preprocessing module

Considering the unstructured information in ESG reports, we optimized the preprocessing procedures to better detect and extract KPIs information from the text and tables from the reports, in order to construct a multitype knowledge base for the following LLM Agent module.

In the initial stage of ESG reports preprocessing, we employed two advanced computer vision tools, including Microsoft's LayoutLMv3 (Huang et al., 2022) and GeoLayoutLM (Xing et al., 2023b), to extract the structural components from the documents, such as headers, paragraphs and tables. We also performed structural analysis by utilizing font characteristics to outline the report's structure, facilitating more efficient information retrieval by linking the outline to the corresponding text. Recognizing the frequent appearance of numerical indicators in tables, we further implemented algorithms like Microsoft's Table Transformer (Smock et al., 2022) and LORETSR (Xing et al., 2023a) to accurately identify table cells and reconstruct table structures, ensuring the precise representation of tabular data.

Following this, we organized all ESG reports into a structured knowledge base, encompassing three components: textual contents, document outlines and table contents. For the textual contents, we use the model mt5 (Raffel et al., 2020) to generate summaries and use the model m3e to create vector representations (Wang et al., 2023). These vector representations were then stored in vector databases like FAISS (Johnson et al., 2019) and Milvus (Wang et al., 2021). Document outlines were treated similarly, with corresponding vectors linked to relevant information for storage. For table contents, we employed a one-to-many mapping to exclude non-essential details, highlighting key information such as ESG indicator names. We then generated vector representations for these keywords, pairing each table with its vector list

Table 1
Number of companies for each of the 12 sectors selected in this study. The full list of companies' names can be found in Supplementary Table 5.

Sector	Number of selected companies
Consumer Staples	14
Properties & Construction	15
Telecommunications	13
Consumer Discretionary	18
Industrials	13
Utilities	12
Financials	11
Energy	16
Healthcare	14
Materials	16
Information technology industry	14
Conglomerates	10
Total	166

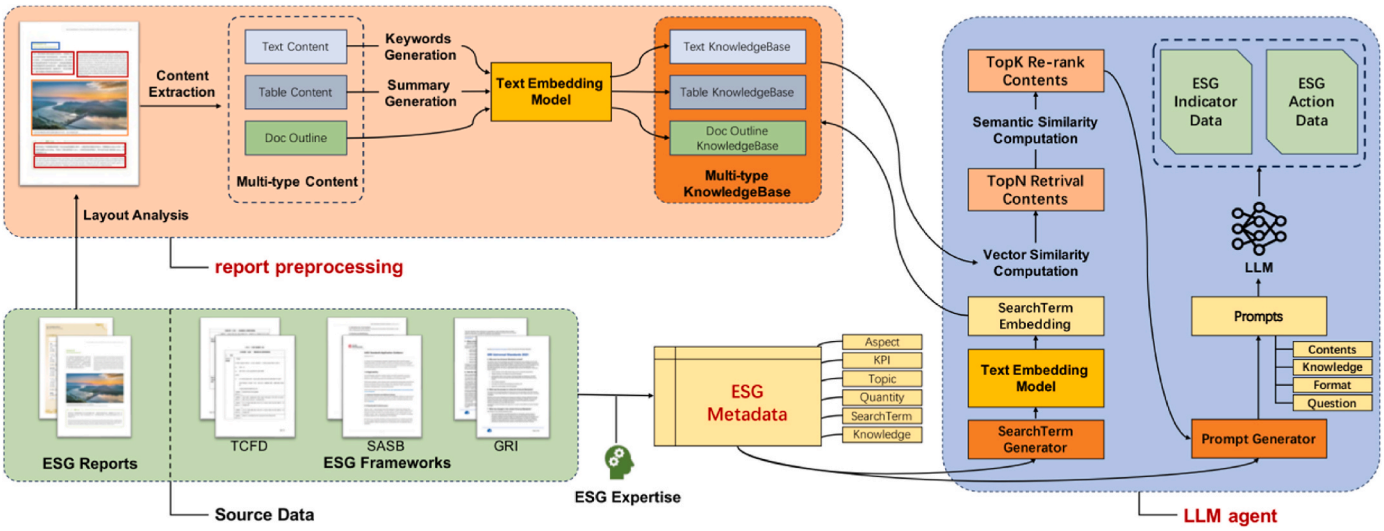


Fig. 1. The three-module design (i.e., the ESG Metadata module, the Report Preprocessing module, and the LLM Agent module) and the workflow of ESGReveal.

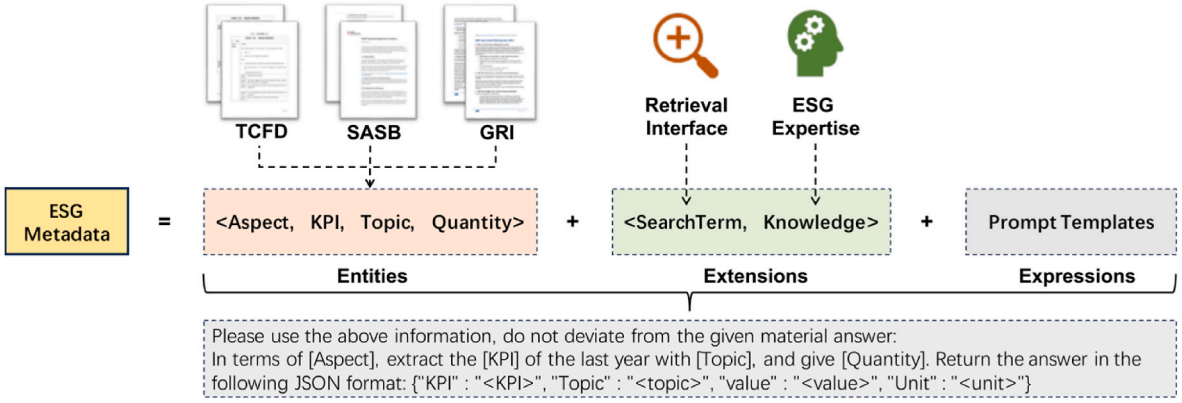


Fig. 2. The three-component structure of the ESG Metadata module: Entities, Extensions, and Expressions.

Table 2
An example of the main contents of one KPI in the ESG Metadata module.

Aspect	KPI	Topic	Quantity
A1. Emissions	Total waste produced (in tonnes) and, where appropriate, intensity (e.g. per unit of production volume, per facility)	[Non-hazardous Waste, Hazardous Waste]	[Absolute Values]

SearchTerm	Knowledge
Construction waste, building rubbish, soil, rubble, organic waste, recyclable fertilizer, paper, plastic bottles, wood, household garbage	Hazardous waste typically includes the following categories: toxic chemical waste, electronic waste, hazardous gas waste, dangerous waste.

to create a streamlined knowledge base for table data.

3.2.3. LLM agent module

Building on the knowledge vector base constructed by the Report Preprocessing module and the ESG Metadata defined by the ESG Metadata module, the LLM Agent module was designed to further retrieve the most relevant ESG information from the constructed knowledge vector base and exporting the final structured answers.

The LLM Agent module generated vectors for the ESG Metadata queries by using the same methods employed as the Report Preprocessing module, calculated the cosine similarity from the ESG Metadata vectors to those in the knowledge vector base, and retrieved the most relevant content from the knowledge base according to the similarity. To improve accuracy and relevance, we applied the coROM (Zhang and Zhang, 2023) model to assess the semantic similarity of the top initial matches, enabling us to refine the ranking. Such a process enabled precise searches for each KPI and pinpoint the most pertinent findings from text, tables, and outlines in the ESG reports.

Based on the ESG Metadata and its most relevant content, we further construct a prompt to extract structural information for each KPI. The prompt comprises the following elements: (1) Preset Information, which provides basic instructions for the LLM's behavior; (2) Reference Content, which includes the retrieved contents from the knowledge base; (3) Expert Knowledge, which incorporates ESG insights from specialists into ESG Metadata module; (4) Question, which frames a targeted query for the indicator crafted from ESG Metadata module, asking about the indicator's disclosure and related data; (5) Answer Format, which presents a structured request for the LLM to report data in the format of <Disclosure KPI, Topic, Value, Unit, Target, Action>, covering disclosure status, ESG performance indicators, topics, numerical values, units,

action targets, and key actions, respectively. Examples of prompt construction and the LLM responses for numeric indicator and textual indicators can be found in Supplementary Table 4.

3.3. Model performance analysis: accuracy assessment for disclosure identification and value extraction

Utilizing ESGReveal, this study extracted both the disclosure status of KPIs and their specific numerical values from sample reports under the HKEx framework. To quantitatively evaluate the accuracy of this extraction process, we calculated two key metrics: Disclosure Identification Accuracy (Acc_{DC}) and Value Extraction Accuracy (Acc_{DE}).

Firstly, we manually annotated each KPI within the sample reports to identify its disclosure status D_{label} and collect their specific numeric values V_{label} by companies. The annotated data is defined as follows:

$$D_{label} = \{d_i \mid i \in \{1, 2, 3, \ldots, N_{mq}\}, d_i \in \{1, 0\}\} \quad (1)$$

$$V_{label} = \{v_i \mid j \in \{1, 2, 3, \ldots, N_v\}\} \quad (2)$$

Here, N_{mq} represents the total number of indicators defined in the ESG metadata module, where the disclosure status for each indicator d_i was marked as either "1" (disclosed) or "0" (not disclosed). Similarly, N_v denotes the total number of numeric indicators actually disclosed, with each specific value v reported.

Next, Disclosure Identification Accuracy (Acc_{DC}) is defined as the ratio of correctly identified disclosure indicators by ESGReveal to the total number of indicators disclosed in the reports; Value Extraction Accuracy (Acc_{DE}) measures the proportion of numeric values correctly extracted by ESGReveal to the number of true numeric values in the sample reports. The formulas for Acc_{DC} and Acc_{DE} are as follows:

$$Acc_{DC} = \frac{1}{N_{mq}} \sum_{i=1}^{N_{mq}} 1(d_i = \hat{d}_i), \quad (3)$$

where $d_i \in D_{label}$ and $\hat{d}_i \in D_{ESGReveal}$

$$Acc_{DE} = \frac{1}{N_v} \sum_{j=1}^{N_v} 1(v_j = \hat{v}_j), \quad (4)$$

where $v_j \in V_{label}$ and $\hat{v}_j \in V_{ESGReveal}$

Here, $D_{ESGReveal}$ and $V_{ESGReveal}$ represent the disclosure status of indicators and specific values extracted by ESGReveal, respectively. These accuracy metrics are calculated as the averaged across the studied samples in order to evaluate the overall performance of ESGReveal.

4. Results

ESG disclosure frameworks encompass a wide range of reporting indicators, broadly categorized into quantitative (e.g., greenhouse gas emissions) and qualitative (e.g., employee diversity policies). Due to difficulties in quantifying the latter, this study focused on the precision of ESGReveal in extracting quantitative indicators as specified by the ESG Metadata Module. We detailed the accuracy assessment method in Section 4.1.1, compared various LLMs' performances in Sections 4.1.2, and discussed ablation study results using the ESG Metadata module framework in Section 4.1.3. This study utilized the ESGReveal tool to dissect selected sample reports, extracting critical information on ESG data disclosures, quantitative figures, and key initiatives, with detailed findings presented across sections 4.2.1 to 4.2.3.

4.1. Assessing performance of ESGReveal

4.1.1. LLM benchmarking

Table 4 shows the performance of ESGReveal in extracting ESG data using different LLMs. Four advanced models were evaluated for comparative analysis: GPT-3.5, GPT-4, ChatGLM (Du et al., 2022), and the QWEN (Bai et al., 2023) model. The experimental results indicate that compared to AccDC, all LLMs generally showed a decline of 4.8% to 7.7%in the data extraction (Acc_{DE}), compared to disclosure coverage (Acc_{DC}). This consistent performance gap may reflect the inherent complexity of the data extraction task, which requires not only identifying the presence of disclosed information but also accurately extracting specific values and attributes. Particularlywhen dealing with the extraction of data on topics such as "employee turnover rate by gender, age group and geographical region" the task involves detailed values for multiple age brackets. This requires that the LLM be capable of accurately processing and distinguishing between multiple data points.

Table 4
Disclosure Identification Accuracy (Acc_{DC}) and Value Extraction Accuracy (Acc_{DE}) across the four LLMs under the ESGReveal framework.

Models	GPT-4	QWEN	GPT-3.5	ChatGLM
Acc_{DE}	76.9%	54.9%	47.1%	46.2%
Acc_{DC}	83.7%	61.4%	51.9%	53.9%

Among the LLMs, GPT-4 achieved the highest accuracy of 76.9%in Acc_{DE} and 83.7% in Acc_{DC}, outperforming the other models. QWEN followed with 54.9% and 61.4% accuracy, respectively. Lastly, GPT-3.5 and ChatGLM performed similarly, with GPT-3.5 achieving 47.1%/51.9%and ChatGLM achieving 46.2%/53.9% accuracy rates. The superior performance of GPT-4 in the analysis of ESG disclosures analysis and indicator extraction can be attributed to its enhanced model capacity and superior comprehension capabilities. In contrast, GPT-3.5, QWEN and Chat-GLM demonstrated comparable performance, though inferior to GPT-4. These findings underscore the pivotal role that a model capacity and interpretive proficiency play in augmenting its effectiveness when applied to specialized domain specific tasks.

4.1.2. Ablation study on the three primary modules

GPT-3.5 and GPT-4 were selected to conduct a series of ablation experiments to evaluate the effectiveness of each module in ESGReveal. A basic RAG implementation was set as a benchmark (LangChainInc, 2023). Under this benchmark, GPT-4 achieved 57.8% accuracy in disclosure analysis and 52.2% in data extraction, while GPT-3.5 achieved 31.7%and 26.6%, respectively.

Building upon the benchmark, we first integrated the Report Preprocessing module and the LLM Agent module, to check if these modules has enhanced the performance of ESGReveal through better document preprocessing and content retrieval methods. As shown in Fig. 3, marked as Enhanced-RAG, GPT-4 saw notable improvements in performance for both of disclosure analysis and data extraction, achieving accuracy of

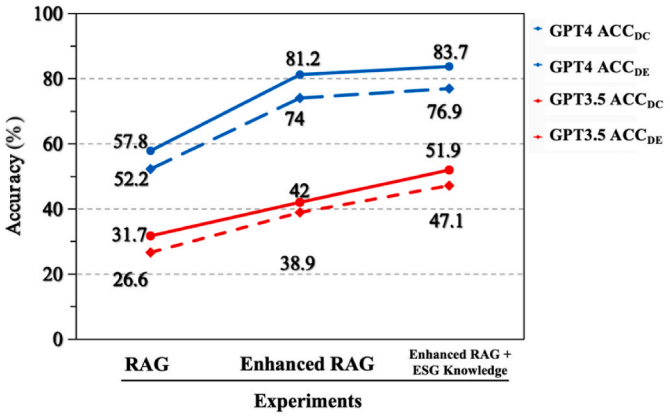


Fig. 3. Ablation study on ESGReveal and performance of data extraction accuracy (Acc_{DE}) and disclosure coverage accuracy (Acc_{DC}) by integrating different modules.

81.2% (+23.4%) and 74.0% (+21.8%) for data extraction. GPT-3.5 also brings improvement of 10.3%and 12.3%, reaching a performance of 42.0% and 38.9% respectively. These results underscore the effectiveness of the Report Preprocessing module and the LLM Agent module proposed in this study. Better structuring of documents and refined retrieval content significantly improve the performance of ESG disclosure analysis and data extraction under the RAG framework.

Next, we checked the effectiveness of the ESG Metadata module. With the introduction of <Knowledge> in the ESG Metadata module, both GPT-4 and GPT-3.5 achieved varying degrees of improvement. GPT-4's accuracy increased to 83.7% (+2.5%) for disclosure analysis and 76.9% (+2.9%) for data extraction, whereas GPT-3.5 saw substantial increases of 9.9%and 8.2%, with accuracy reaching 51.9% and 47.1%, respectively. This indicates that by incorporating supplementary knowledge from the ESG domain, LLMs can experience different levels of enhancement in their parsing abilities. Especially for the LLMs with weaker model capacity and comprehension abilities (e.g., GPT-3.5), the accuracy gains are much greater than those for LLMs with stronger capacities and comprehension (e.g., GPT-4). This implies that the addition of ESG domain knowledge in ESG Metadata can greatly enhance the performance ofless capable LLMs in ESG disclosure analysis and data extraction tasks.

4.2. Dissecting ESG reports: disclosure, metrics, and key actions

4.2.1. Disclosure degree

In this study, 34 numerical KPIs are designed in the ESG Metadata according to the HKEx framework: 12 for environmental dimension and 18 for social dimension. We found that, most listed companies in HKEx primarily report data on these two dimensions in their ESG reports. Following the practice of professional ESG rating agencies like S&P Global and MSCI, which categorize ESG rating results into high, medium, and low tiers based on score ranges (S&P Dow Jones Indices, 2023), and in conjunction with the actual disclosure practices with industries, we adopted a three-zone assessment system for evaluating disclosure levels of each ESG report. These three zones are defined by the disclosure level as green (over 80%), yellow (between 60% and 80%), and red (below 60%).

Fig. 4 displays the disclosure rate of environmental, social, and overall numerical indicators across various sectors. It reveals an average environmental disclosure rate of 69.5% (yellow zone), while 57.2% (red zone) for the average social disclosure, indicating that the listed companies generally disclose better on environmental indicators rather than social indicators. Moreover, even among the representative subset of large capital companies listed on the HKEx, no sector achieved an average disclosure level exceeding 80%. This highlights the urgent need to strengthen ESG disclosure practices across sectors.

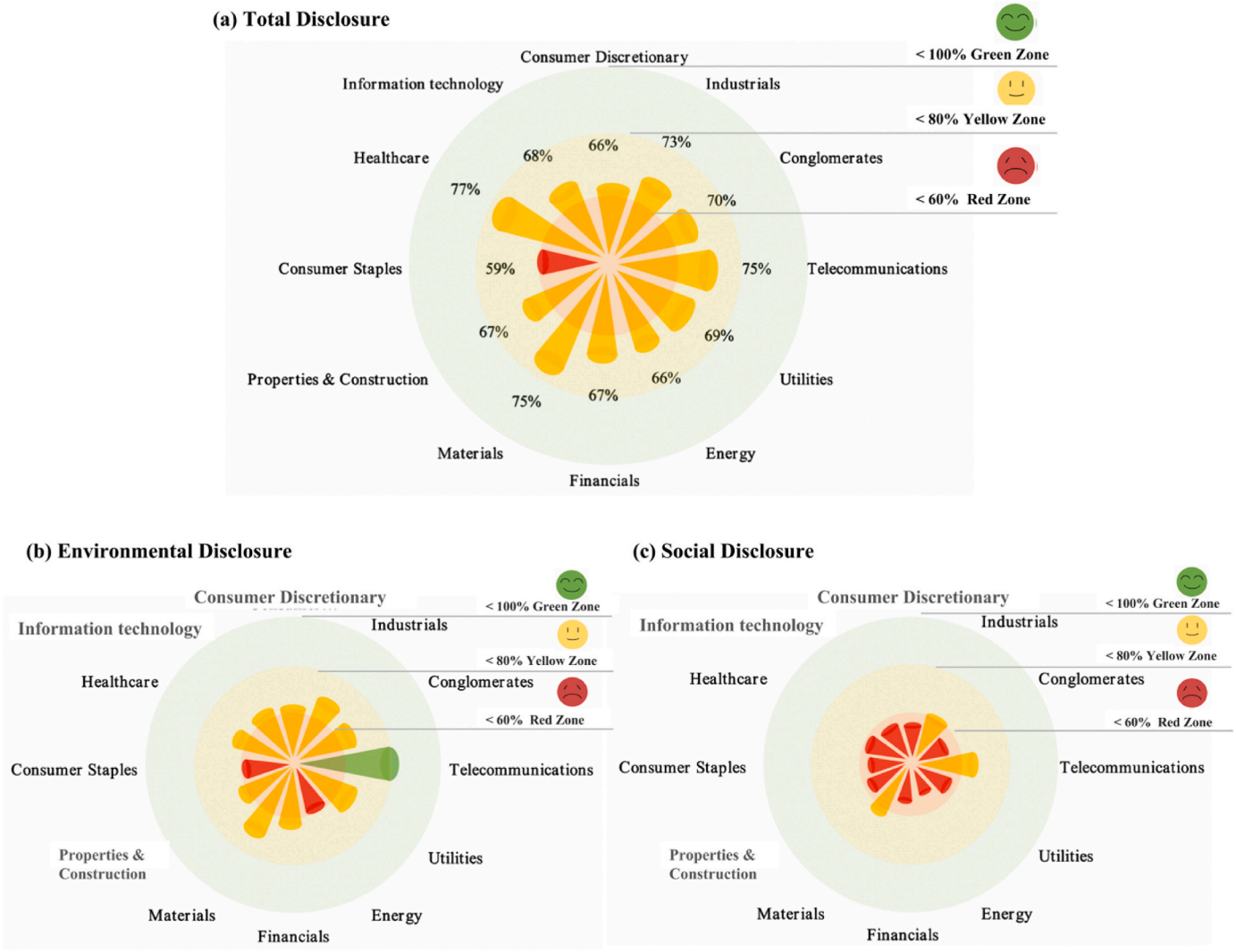


Fig. 4. Disclosure levels by sector: (a) overall, (b) on environmental issues, and (c) social issues.

From an industry perspective, sectors such as telecommunications (e.g., China Tower Corporation Ltd., China Mobile Ltd., and China Unicom [Hong Kong] Ltd.), healthcare (e.g. China Biologic Products Holdings Inc., and Alibaba Health Information Technology Ltd.), and materials (e.g. Northern Mining Ltd) exhibit higher disclosure rates. In contrast, the consumer staples sector (e.g., China Starch Holdings Ltd., and Want China Holdings Ltd.) and the energy sector (e.g., China National Offshore Oil Corporation Ltd., and China Petroleum & Chemical Corporation Ltd.) demonstrate weaker performance. Upon further examination of the healthcare industry, which exhibits the highest disclosure level, we found that five companies were in the green zone, while the remaining six were in the yellow zone (see Supplementary Fig. 1).

4.2.2. Numeric metrics

In analyzing ESG reports with ESGReveal, we not only examined the disclosure status but also extracted quantitative data. For instance, we collected data on Scope 1 (direct) and Scope 2 (indirect) emissions. To account for industry differences and company sizes, we normalized company market values and calculate greenhouse gas emission intensity per million Hong Kong dollar of market value. This approach enabled us to make emission data comparable across companies.

As shown in Fig. 5, we observed a general trend in which Scope 2 emissions were higher than Scope 1 emissions across most sectors. However, certain sectors, such as utilities, energy, healthcare and materials, exhibited higher Scope 1 emissions. The utilities sector, in particular, reported elevated Scope 1 emissions due to its inclusion of energy-intensive enterprises, such as natural gas and electricity providers. In contrast, the financials sector, being a typical labor-intensive industry, recorded higher Scope 2 emissions data, reflecting its significant reliance on electricity.

4.2.3. Key actions

Our analysis of key actions, extracted from ESG reports, provided valuable insights into core industry pratices. Supplementary Fig. 2 highlights the five most frequently mentioned terms in cross-industry discussions on environmental issue. For example, in the context of greenhouse gas emissions, “greenhouse gas reduction” was the most frequently mentioned phrase, appearing in 24.3% of reports. Similarly, for water resource management, the terms “adopt water saving practices” and “water saving training for staff” are each mentioned in over 10%. Regarding fossil fuel usage, the data indicate a gradual shift towards renewable energy, as evidenced by terms like “renewable energy” and “green transition”. In discussions on hazardous and non-hazardous waste management, terms like “pollution source control” and “compliant emissions” were prevalent. Additionally, topics related to electricity and energy usage highlighted “green office practices” and “smart energy control”. Overall, the comprehensive analysis of high frequency vocabulary suggests that phrases such as "ESG management" and "green transition" appear consistently in ESG reports across industries, reflecting common efforts to reduce environmental impact.

Supplementary Fig. 2 lists key action terms that are closely associated with specific sectors. For instance, in the properties and construction sector, represented by companies like China National Building Material Company Ltd. and China Resources Cement Holdings Ltd., there is a strong focus on controlling dust and volatile organic compound emissions, implementing water-saving systems, and reusing building materials. In the telecommunications sector, companies such as China Tower Corporation Ltd. and China Mobile Ltd. emphasize research and development of environmentally friendly products, advancing battery recycling and reuse, and enhancing environmental monitoring and pollution control related to communication towers. As for the healthcare sector, companies like China Biologic Products Holdings, Inc. and Alibaba Health Information Technology Ltd. are prioritizing the promotion of digital medical services as their ESG efforts. financials sector, meanwhile, plays a leading role in green finance, utilizing financial instruments to support the sustainable development of various industries.

5. Discussion

This study introduces ESGReveal, a methodology for extracting structured data from ESG reports using advanced LLMs and RAG techniques. Our approach shows several improvements over existing research: 1. Integration of LLM and RAG: Unlike prior studies that have utilized single NLP technology, such as BERT by Ruberg et al. (2023), ESGReveal integrates LLMs with RAG, enhancing the precision and efficiency of data extraction. 2. Structured Data Extraction: Previous works, including those by Luccioni et al. (2020) and Fischbach et al. (2022), have focused on qualitative analyses. In contrast, ESGReveal emphasizes the structured extraction of both numerical and textual data, providing a more comprehensive dataset for analysis. 3. Comprehensive ESG Metadata Module: The introduction of the ESG metadata module, as detailed in Section 3.2, allows for a systematic approach to ESG data extraction that is not present in other tools, such as the ESG-Miner by Fischbach et al. (2022). 4. Customizability and Adaptability: Our methodology offers a higher degree of customizability and adaptability across various ESG reporting standards, which has been a noted limitation in studies like that of Pasch and Ehnes (2022).

Although this study has yielded some initial findings, we have identified several factors that suggest avenues for furtherresearch. First, we have noted substantial differences in the capabilities of various LLMs when it comes to ESG analysis tasks. These differences underscore the reliance of LLMs on theirintrinsic computational abilities and the depth of their ESG domain knowledge when performing specialized analyses. Second, the structural parsing and retrieval accuracy of ESG reporting documents are of paramount importance when LLMs are employed for ESG disclosure analysis. Enhanced document structuring and more effective information retrieval can notably improve the performance of the RAG framework in terms of data extraction and analysis. Lastly, given that key information in ESGreports is sometimes presented in pictorial forms, our current ESGReveal approach falls short in extracting such data. We aim to rectify this in our future work by refining and

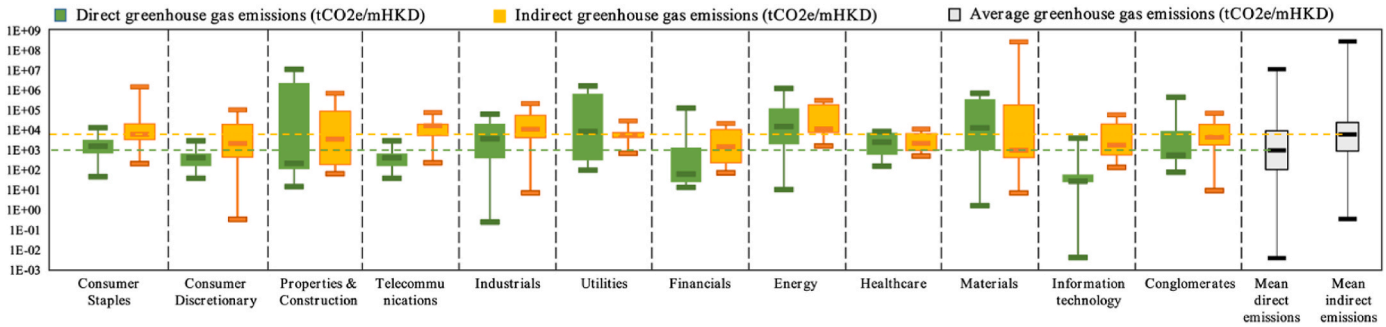


Fig. 5. Comparative analysis of greenhouse gas emission intensity for direct and indirect emissions by industry.

optimizing our approach to accommodate these kinds of information.

Looking to the future, research in this domain can evolve in several promising directions. One key direction is the meticulous development of more granular, industry specific ESG datasets. Such datasets can drive highly accurate data analyses, significantly boosting the transparency and credibility of corporate ESG disclosures. Moreover, the automated extraction techniques pioneered in this study are anticipated to broaden their reach, with potential applications extending to the analysis ofpivotal documents such as those from the Inter-governmental Panel on Climate Change. This expansion would leverage digital technology to make a meaningful impact on critical global conversations around climate change mitigation and sustainable development.

In summary, the observed gaps in environmental and social data disclosure highlight the ongoing challenges companies face in achieving transparency. This suggests a need for stronger regulatory frameworks and industry standards to guide more consistent and comprehensive ESG reporting. Additionally, the accuracy and effectiveness of the ESGReveal framework in extracting structured data from unstructured reports could have significant implications for how organizations approach ESG disclosures in the future. By automating and standardizing the extraction process, companies can enhance their reporting accuracy and efficiency, potentially leading to more informed decision making by investors and stakeholders. Future research should explore the scalability of ESGReveal across different industries and geographical regions, which could further validate its utility and drive global adoption.

6. Conclusion

This study presents ESGReveal, a novel methodology for the systematic extraction and analysis of ESG data from reports. Such a three-module-design framework of ESGReveal demonstrates the utility of automated methodologies in processing unstructured ESG reports, significantly enhancing the accuracy and efficiency of ESG disclosure assessments. By incorporating the Preprocessing Report module and the LLM Agent module, the accuracy of disclosure identification and value extraction tasks improve by over 20% for GPT-4 compared to the baseline performance. Furthermore, the introduction of the ESG Metadata module enhances the accuracy of disclosure identification and value extraction by an additional 2.5% and 2.9%, achieving overall accuracies of 83.7% and 76.9% in the HKEx ESG report analytical task.

Applying ESGReveal, we analyzed the 2022 ESG reports of 166 top-listed companies across 12 sectors on the HKEx. Following the HKEx disclosure guidelines, we identified and extracted the environmental and social KPIs. The average disclosure rate for the 59 environmental and social KPIs across the 12 sectors was 69.3%, with environmental KPIs having a higher disclosure rate (69.5%) compared to social KPIs (57.2%). None of the 12 sectors achieved an average disclosure rate exceeding 80%. The Healthcare sector had the highest average disclosure rate at 77%, while the Consumer Staples sector had the lowest at 59%. For environmental KPIs, the Telecommunications sector showed the highest disclosure rate, while only Telecommunications, Industrials, and Materials sectors exceeded a 60% disclosure rate for social KPIs.

In conclusion, the ESGReveal framework we proposed presents a sufficient and effective tool for analyzing corporate ESG performance from unstructured ESG reports. Such a framework has demonstrated its capability to perform automated and accurate ESG performance analysis, and future work will focus on expanding its scope to cover a larger database of corporate ESG metrics. On one hand, ESGReveal will enable continuous tracking of companies' sustainability actions across environmental and social domains. On the other hand, it will help identify and highlight successful sustainability policies, providing a valuable reference for other companies to adopt best practices. Furthermore, ESGReveal will offer regulators, e.g., HKEx, a powerful tool to monitor whether listed companies have complied with mandatory ESG disclosure requirements or provided sufficient explanations. This tool can also be expanded to other contexts, such as extracting information from the submission of biennial transparency reports by countries under the Paris Agreement. These extracted information, which include updated national greenhouse gas inventories and mitigation actions, will facilitate the global stocktake process under the United Nations Framework Convention on Climate Change (UNFCCC).

CRediT authorship contribution statement

Yi Zou: Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Mengying Shi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Zhongjie Chen:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhu Deng:** Supervision, Resources. **Zongxiong Lei:** Resources. **Zihan Zeng:** Resources. **Shiming Yang:** Resources. **Hongxiang Tong:** Resources. **Lei Xiao:** Resources. **Wenwen Zhou:** Supervision, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author (s) used GPT-4 in order to improve readability and language. After using this tool, the author (s) reviewed and edited the content as needed and take (s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2024.144572.

Data availability

Data will be made available on request.

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